

Foundation (Jalapeno)

- Q1.** What is the purpose of a stem-and-leaf plot in data analysis?
- (A) To compare two sets of data
(B) To display the distribution of a dataset
(C) To calculate the mean of a dataset
(D) To create a scatter plot
(E) To find the median of a dataset
- Q2.** A space experiment measured the temperature of a planet in $^{\circ}\text{C}$. The results were: 25, 27, 28, 29, 30, 31, 32. Which stem-and-leaf plot represents this data?
- (A) 2 | 5, 7, 8, 9, 0, 1, 2
(B) 2 | 5, 7, 8, 9, 0, 1
(C) 2 | 0, 1, 2, 5, 7, 8, 9
(D) 3 | 0, 1, 2, 5, 7, 8, 9
(E) 2 | 0, 1, 2, 3, 5, 7, 8
- Q3.** In a stem-and-leaf plot, what does the 'stem' represent?
- (A) The ones digit of a number
(B) The tens digit of a number
(C) The hundreds digit of a number
(D) The thousands digit of a number
(E) The decimal part of a number
- Q4.** A lab experiment measured the pH levels of 12 different solutions. The results were: 4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.2, 7.5. Which of the following is the correct stem-and-leaf plot for this data?
- (A) 4 | 2, 5, 8, 5 | 1, 4, 7, 6 | 0, 3, 6, 9
(B) 4 | 2, 5, 8, 5 | 1, 4, 7, 6 | 0, 3, 6
(C) 4 | 2, 5, 8, 5 | 1, 4, 7
(D) 5 | 1, 4, 7, 6 | 0, 3, 6, 9
(E) 6 | 0, 3, 6, 9, 7 | 2, 5
- Q5.** What is the primary advantage of using a stem-and-leaf plot?
- (A) It can display multiple datasets at once
(B) It can calculate the mean and median of a dataset
(C) It can display the distribution of a dataset
(D) It can compare two or more datasets
(E) It can create a box-and-whisker plot
- Q6.** A dataset has the following stem-and-leaf plot: 3|1, 4, 7. What are the actual values in the dataset?
- (A) 31, 34, 37
(B) 30, 31, 34
(C) 3.1, 3.4, 3.7
(D) 13, 14, 17
(E) 1, 4, 7
- Q7.** Which of the following is NOT a characteristic of a stem-and-leaf plot?
- (A) It displays the distribution of a dataset
(B) It can compare multiple datasets
(C) It is used to calculate the mean of a dataset
(D) It displays the actual values in a dataset
(E) It can create a box-and-whisker plot
- Q8.** What is the difference between a stem-and-leaf plot and a histogram?
- (A) A stem-and-leaf plot displays the actual values, while a histogram does not
(B) A stem-and-leaf plot is used for categorical data, while a histogram is used for numerical data
(C) A stem-and-leaf plot is used for small datasets, while a histogram is used for large datasets
(D) A stem-and-leaf plot displays the distribution of a dataset, while a histogram displays the frequency of a dataset
(E) A stem-and-leaf plot is used for numerical data, while a histogram is used for categorical data

Open Ended Questions (FRQ)

- Q9.** The following dataset represents the masses of 10 different asteroids in grams: 100, 120, 150, 180, 200, 220, 250, 280, 300, 320. Create a stem-and-leaf plot for this dataset and describe the distribution of the masses.
- Q10.** A space agency collected data on the orbital periods of 8 different planets in our solar system. The data is as follows: 10.2, 12.5, 15.8, 18.1, 20.4, 22.7, 25.0, 27.3. Create a stem-and-leaf plot for this data and interpret the distribution of the orbital periods.

Chef's Tips

- Tip 1: Make sure to check for any missing values in the dataset
- Tip 2: Use the correct scale when creating a stem-and-leaf plot
- Tip 3: Pay attention to the units of measurement when interpreting the distribution of a dataset